Scalability & Future Plan

Date	3 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

The **Emotion Detection and Learning Support Engine** is designed to accurately detect users' emotions from text and provide personalized learning support through an AI-powered web application. While the current system successfully meets the project objectives, there are several opportunities to improve its scalability, performance, security, and user experience. Future enhancements will focus on supporting a larger number of users, improving the emotion detection model, integrating advanced AI technologies, and deploying the application on a cloud platform for real-world use.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses a limited emotion dataset for model training.	May reduce prediction accuracy for diverse real-world user inputs.	High
2	Runs only on a local Streamlit server.	Limits access for multiple users and remote availability.	High
3	Basic user interface without authentication	Suitable only for project demonstration and testing.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users	Deploy the application on a cloud platform such as Streamlit Community Cloud, AWS, or Azure to support multiple concurrent users.

2	Data Storage	Uses local files for storing data and models.	Integrate a relational database such as MySQL or PostgreSQL to manage user information and learning history efficiently.
3	Performance	Uses a locally stored BERT model for prediction	Optimize the model using quantization or model compression and deploy on a high-performance cloud server for faster predictions.
4	Security	Basic input validation	Implement user authentication, secure login, encrypted data storage, and secure API communication to protect user information.

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	Integrate user authentication, database support, and user progress tracking.	Next 3 Months	Secure user access and personalized learning experience.
Phase 3	Add multilingual emotion detection, and AI-powered chatbot support for learning guidance.	Next 6 Months	Improve accessibility and provide interactive learning assistance.
Phase 4	Deploy the application on a cloud platform with a responsive mobile interface and real-time analytics dashboard.	Next 12 Months	Support large-scale users and improve accessibility, and enable real-world deployment.